

EthernaCare

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[Ethernacare]

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Declaration

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Abstract

EthernaCare is a Flutter mobile application designed to support elderly people and independent individuals who live alone. It combines a daily virtual-pet check-in, configurable inactivity monitoring, trusted-contact escalation, location-assisted emergency records, rewards, legacy planning, weather-responsive visuals, and AI guidance. The objective is to reduce delayed awareness of prolonged inactivity while providing an accessible way to organize safety information and personal preferences.

The daily safety signal is completed by tapping Oren, the virtual cat. Each successful daily check-in is stored in Supabase and can award Oren tokens and streak-based rewards. The Android app uses WorkManager for best-effort periodic inactivity checks. A local reminder counter progresses through three missed threshold windows; reminders 1 and 2 prompt the user, while reminder 3 starts the configured escalation. When the primary trusted contact option is selected, the app attempts an automated SMS and records the alert and available location. The app never automatically calls Malaysia's 999 service; a 999 call requires an explicit user action.

The project follows an Agile approach. Flutter implements the responsive Android and web interface, while Supabase provides authentication, PostgreSQL data storage, Row Level Security, private file storage, and Edge Functions. Malaysia's official weather API is used first, with Open-Meteo as a location-based fallback. Gemini provides server-side AI guidance with a built-in offline fallback. SharedPreferences caches selected dashboard, weather, reward, chat, Oren, and inactivity state to reduce repeated network requests. Automated widget and logic tests verify the principal user flows, while external SMS, OAuth, weather, and AI providers require configured credentials and integration testing.

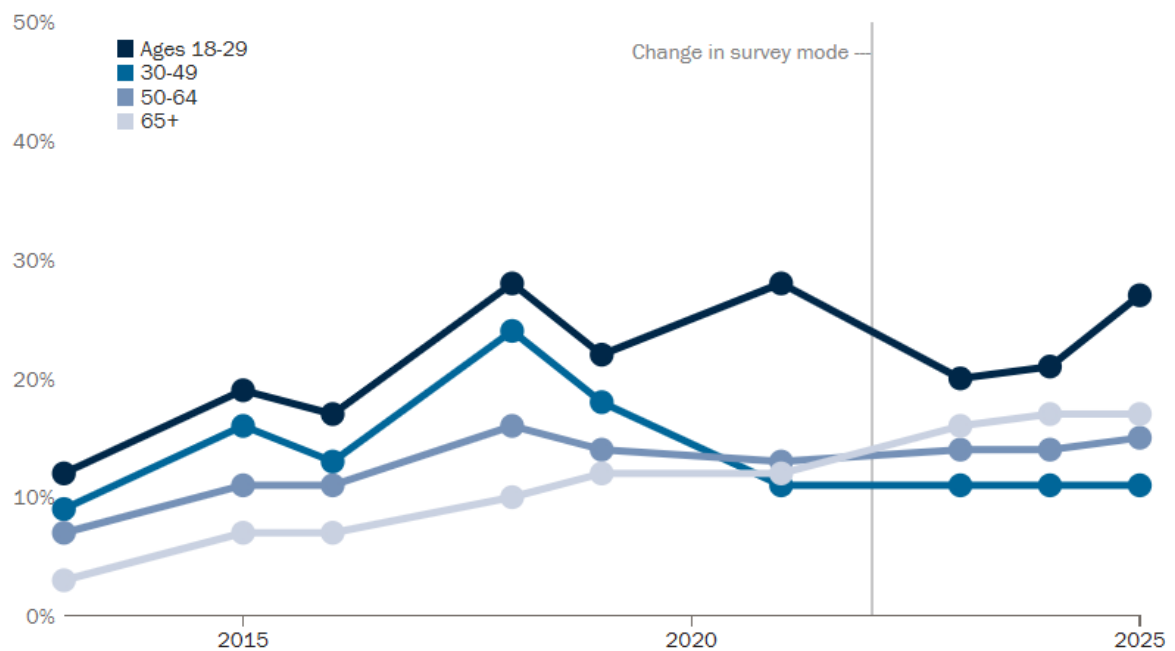
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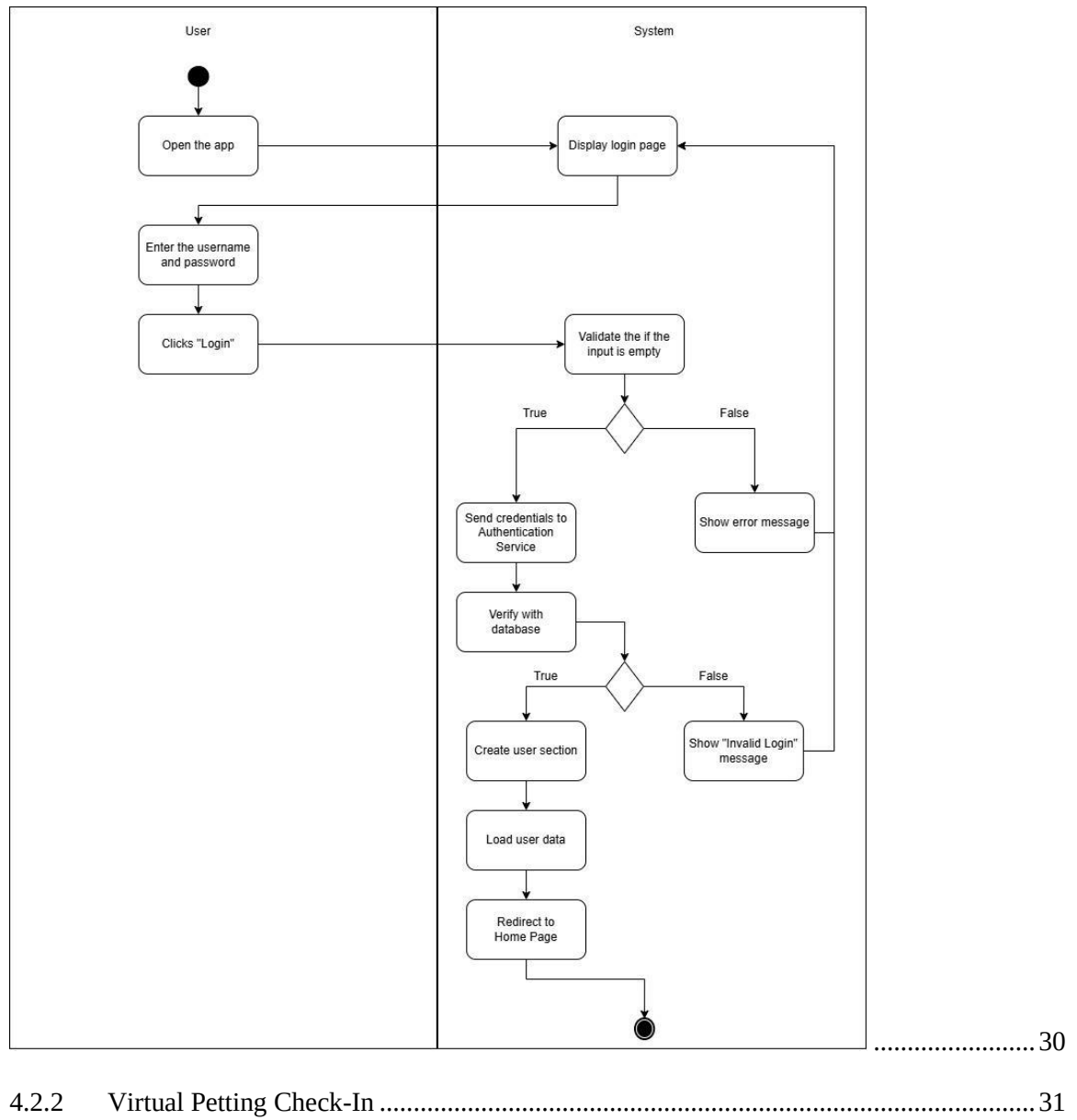
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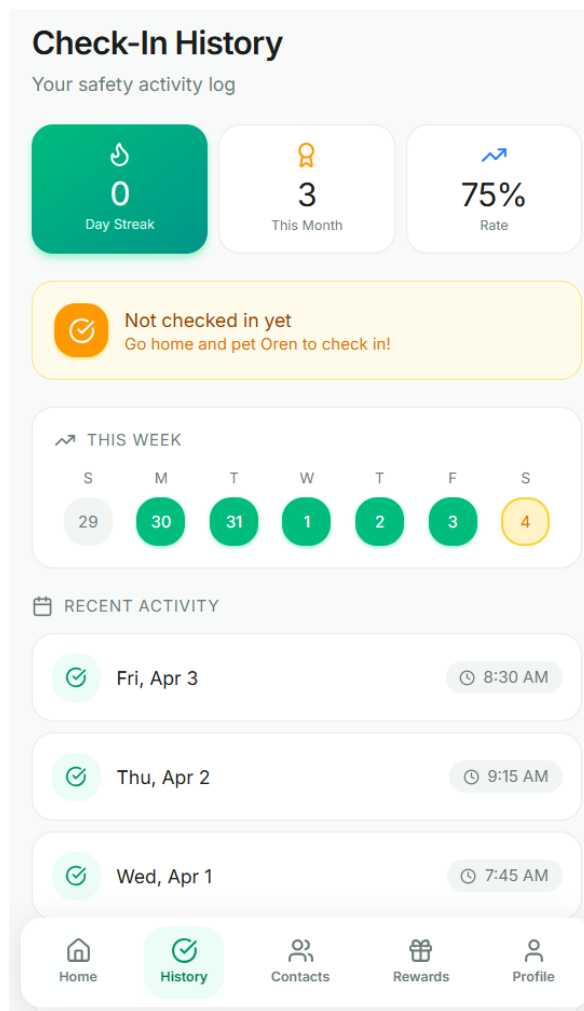
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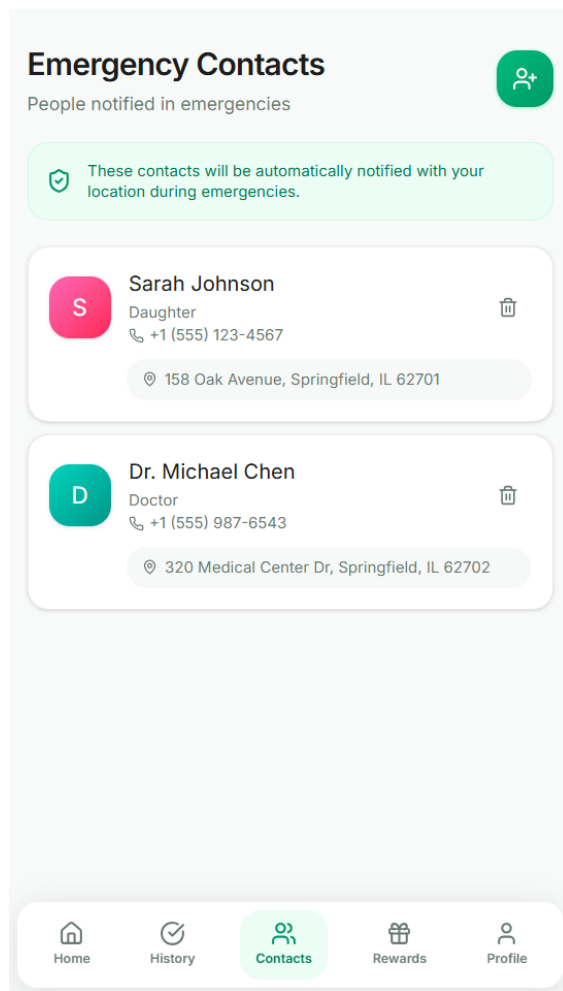


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Chapter 1

Introduction

1 Introduction

This chapter presents the background and motivation of the Ethernacare project. It explains the social problem related to elderly individuals living alone and the risks linked to delayed discovery during emergencies. The chapter also outlines the project objectives, scope, and significance to provide a clear understanding of what the system aims to achieve.

1.1 Project Objective(s)

The project aims to utilize a **gamified interaction** such as **petting a virtual cat** to confirm elderly users are active. Instead of traditional messages, the user interacts with a virtual companion to record a check-in, making the safety verification process more engaging and less clinical.

Another objective of the project is to support elderly users in preparing for post-death matters. This includes allowing users to record their funeral preferences and store digital copies of wills in a secure manner. By providing this function, the project helps reduce confusion, emotional stress, and administrative difficulties for family members after the user's death.

The project also aims to provide basic information support through an AI-assisted chatbot. This feature is designed to help users and family members access general guidance related to funeral arrangements and related matters in a simple and direct way.

Overall, the project fulfills the requirements of “SDG 3 – Good Health and Well-Being” and seeks to provide a practical and accessible solution that improves elderly safety, preserves personal dignity, and supports families by using smartphone technology that users already own.

1.2 Background of the Project

1.2.1 Malaysia is about to become an aging society

According to the prediction from “Portal Rasmi Kementerian Kewangan”, Malaysia will become an aged nation in 2048 (Azizan, 2025). The number of elderly people living alone will definitely increase as the elderly population increases. Moreover, many elderly people have poorer health compared to younger people and face a higher risk of death. These conditions increase the chance of dying alone without immediate discovery.

1.2.2 Migration and Elderly Parents Living Alone

In recent years, many young adults have moved from home to other areas for work or education. Internal migration remained dominant at 63.5% in 2024 (up from 62.3% in 2022), Urban to urban migration dominated at 84.6% in 2024 (up from 79.3% in 2022)(*Migration Survey Report, Malaysia, 2024, 2025*). This often happens when the children are grown up and want to start their own lives. As a result, parents are left living alone in their house. The contact and interaction between parents and children becomes less frequent, children are busy in their own business and usually only back to home on certain holidays. When elderly parents live alone, they are facing the higher risks of dying alone because help may not be available immediately. This situation increases social isolation.

1.2.3 Increase in the Number of Single Individuals

The number of marriages decreased 12.5 per cent from 215,022 (2022) to 188,100 (2023). (*Migration Survey Report, Malaysia, 2024, 2025*). In 2020, there were 14.5 male and 14.3 female births per thousand population, a decrease from 15.1 male and 14.9 female births per thousand population in 2019 (*International Journal Of Academic Research In Business & Social Sciences, 2024*). It was the lowest rate recorded in the country. Combined with these factors, the number of single individuals without children will increase in the future. Against the backdrop of soaring consumption levels, many people are facing delayed marriage, financial issues, work pressure or personal choices. Others remain single due to divorce or the loss of a spouse. As a result, more adults live alone for longer periods of time. This change affects daily support systems, especially as people grow older. Decades later, the number of living alone elderly people will inevitably increase.

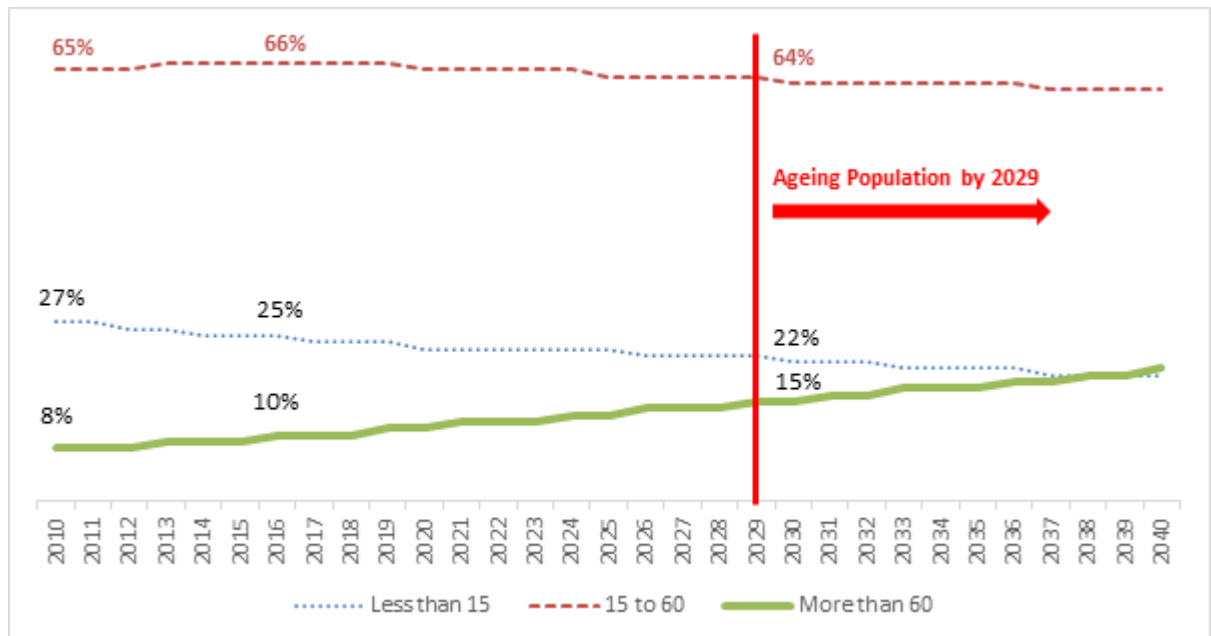


Figure 1.2.1: The prediction of the population age group in the future

(Malaysia Population Research Hub, n.d.)

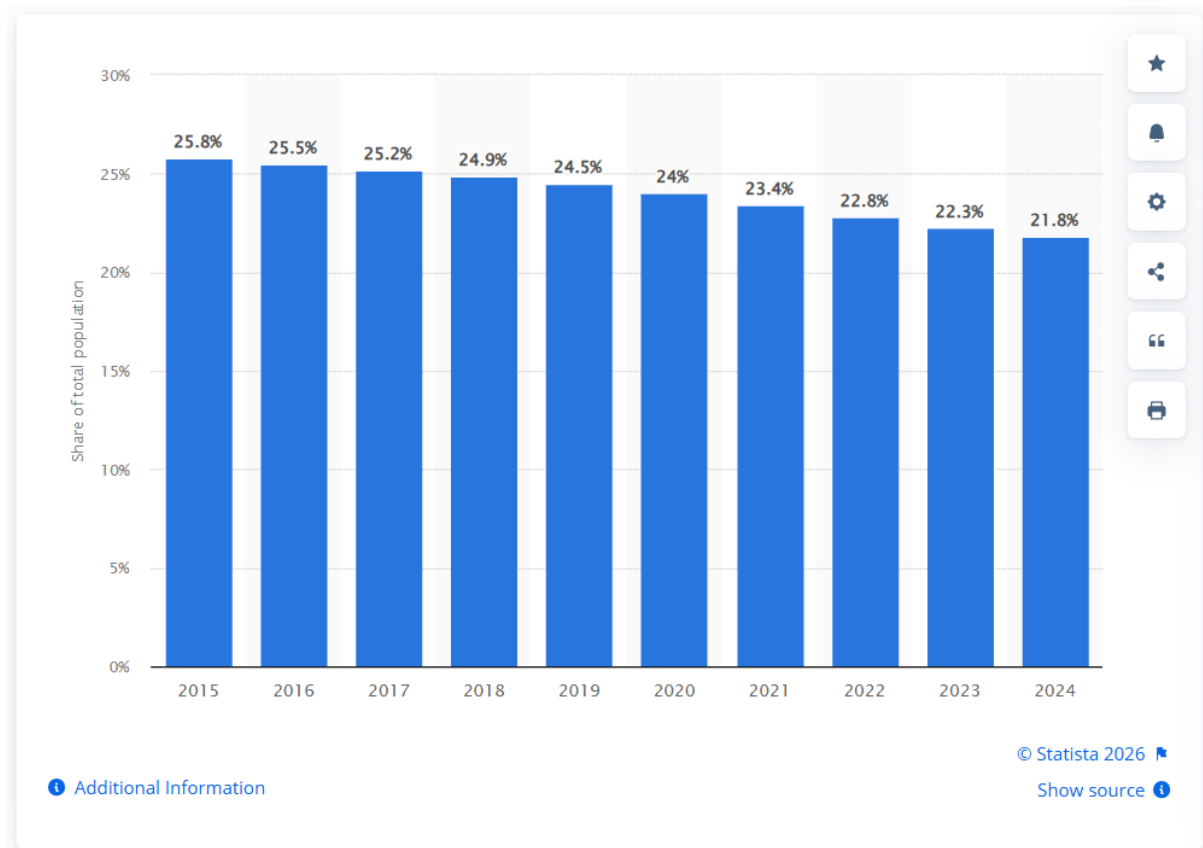


Figure 1.2.2: Children in total population in Malaysia from 2015 to 2024

(Statista, 2025)

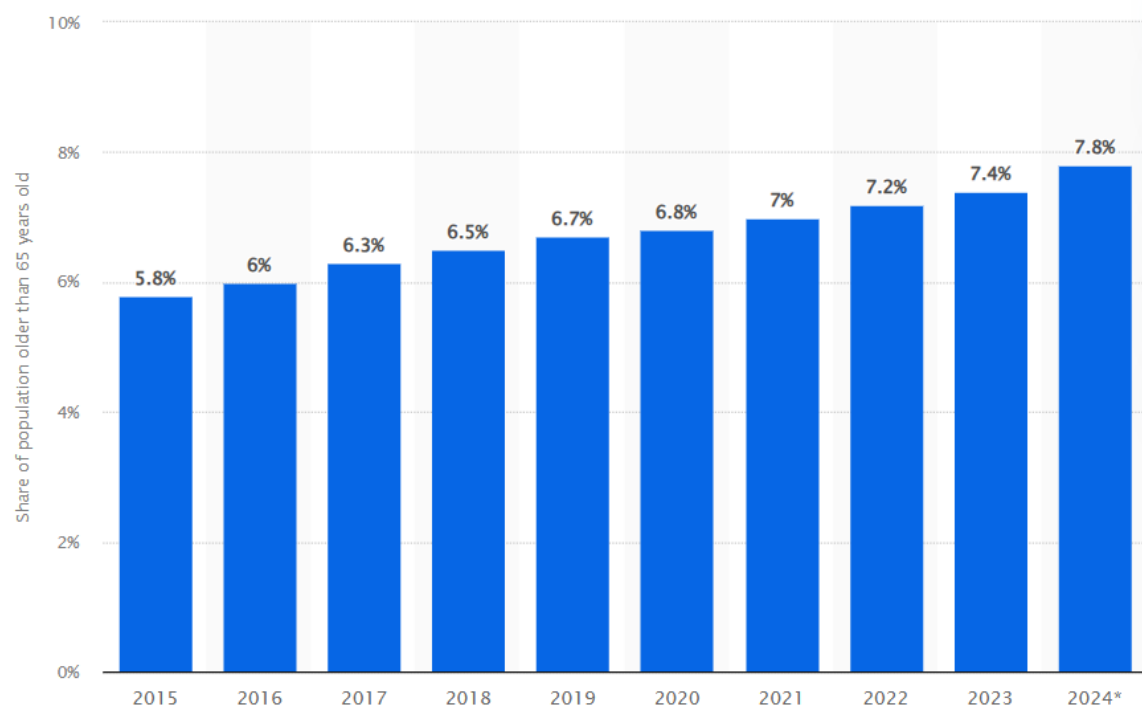


Figure 1.2.2: Aging population Malaysia 2015-2024

(Statista, 2025)

1.3 Advantages & Contributions

1.3.1 Easy to access and achieve

The project provides a solution that does not require extra devices or complicated devices. It relies on the smartphones that are owned by the users. The system uses a **gamified pet-care interaction** that encourages regular use. By transforming the check-in into the act of 'caring' for a virtual cat, the system reduces the 'surveillance' feel of the app and replaces it with a sense of companionship and purpose.

1.3.2 Reducing the unfortunate cases & Notified families in time

The project contributes by reducing delays in emergency response and death discovery through **daily check-in verification**. Once the user response is out of time, user families will auto receive the notification. Even though the elderly user cannot be rescued in time, at least can be found in time to prevent the body from rotting.

1.4 Project Plan

The project follows an Agile mobile application development approach. Development is divided into several phases that include planning, system design, implementation, testing, and deployment. Core features such as activity detection and emergency alert notification are developed early in the project. Features related to funeral preferences, will management, and AI-assisted guidance are developed in later stages. Testing is carried out throughout the development process to ensure that each function works correctly and meets user needs.

Project Schedule:		
ACTIVITIES	EXPECTED OUTCOME	COMPLETION DATE
Proposal	Determine the objective, problems and solutions.	16/12/2025
Phase 2: FYP1		
Chapter 1: Introduction	Introduce the project purpose, background, objective and justification	26/1/2026
Chapter 2: Research background	Retrieve the relevant information, system requirements and research gaps	7/2/2026
Chapter 3: Methodology	Defined the system development tools and requirements.	24/2/2026
Chapter 4: System Design	Mock up prototype, system architecture and data flow diagram	14/3/2026
Project I Portfolio	Submission	13/4/2026
Phase 3: FYP2		
Chapter 5: Development & Testing	Develop the main functions	26/6/2026
Chapter 6: System deployment	Deploy and test the system in real-world scenarios.	6/7/2026
Chapter 7: Project conclusion	Final summary and outcome of the project	1/8/2026

Figure 1.4: The project schedule for each phases

1.5 Project Team & Organization

This project is developed as an individual final year project under the guidance of a project supervisor. The student is responsible for system design, development, testing, and documentation. The supervisor provides academic guidance, technical advice, and feedback to ensure that the project meets academic requirements and follows proper development practices.

1.6 Chapter Summary and Evaluation

This chapter introduced the Ethernacare project, highlighting its shift from traditional monitoring to a **gamified wellness check-in system**. By replacing standard greeting messages with a **virtual pet interaction (petting a cat)**, the project aims to increase user engagement and provide a more intuitive "heartbeat signal" for elderly safety. The overall plan and objectives were defined to ensure this gamified approach effectively addresses the risks of delayed emergency discovery.

Chapter 2

Literature Review

2 Literature Review

This chapter examines previous research and existing systems related to elderly individuals who live alone and the growing concern of lonely deaths. It looks at demographic changes such as population ageing and the rise in independent living among older adults. It also reviews current technology solutions, including wearable emergency devices, smart home monitoring systems, and mobile safety applications. Through this review, the chapter highlights the strengths and weaknesses of existing approaches and clarifies how the proposed project addresses identified gaps and limitations.

2.1 Aging Population in Malaysia

2.1.1 Demographic Shift Toward an Ageing Society

Malaysia is moving toward an ageing nation as the percentage of citizens aged 60 and above continues to increase. The Department of Statistics Malaysia (*Current Population Estimates, Malaysia, 2023*, 2023) projects that older adults will make up around **15%** of the population by 2030. This change is mainly influenced by longer life expectancy and lower birth rates. The transition creates new challenges in healthcare planning, elderly care services, and community support systems.

2.1.2 Increase in Elderly Living Alone

The increase in elderly individuals living alone is closely linked to broader demographic and social transitions. According to the *World Report on Ageing and Health* by the World Health Organization (2015), although more generations within a family may now be alive at the same time due to increased life expectancy, these generations are increasingly living separately. In many countries, the proportion of older people living alone is rising dramatically. For example, in several European countries, more than 40% of women aged 65 and above live alone. Even in societies with strong traditions of multigenerational living, such as Japan and India, extended family households are becoming less common. Declining fertility rates mean there are fewer younger family members available to provide care, while changing gender roles have reduced the traditional caregiving capacity within families. Smaller household sizes may limit reciprocal care arrangements and increase the risk of poverty. Furthermore, older adults living alone may face higher risks of social isolation and suicide. Although many older people prefer to remain in their own homes and communities for as long as possible, these structural changes suggest

that traditional family-based care models are becoming less sustainable in the modern era (World Health Organization, 2015).

2.2 Technology concepts

2.2.1 Smartphone-Based Passive Monitoring

Smartphone-based monitoring systems use built-in sensors and activity logs to observe user behavior without requiring constant manual input (Wang, 2014). This passive approach allows the system to detect unusual patterns such as prolonged inactivity, which may indicate potential emergencies. Because the monitoring process runs automatically in the background, it reduces the need for continuous interaction and can operate without disturbing users.

2.2.2 Daily Check-In Monitoring Systems

Daily check-in monitoring systems are used in safety and well-being applications to verify that users are active through simple interactions with a mobile device. Instead of continuously collecting sensor data, gamified check-in systems utilize **virtual pets or interactive elements** to maintain user engagement. By requiring a tactile interaction like 'petting' an on-screen character, the system ensures the user is physically capable and responsive, serving as a more interactive form of passive monitoring (Rashidi & Mihailidis, 2013).

2.2.3 Wearable or Sensor Monitoring

Wearable and sensor-based monitoring systems are commonly used in elderly care technologies to continuously track health conditions and daily activities. These systems typically involve wearable devices such as smartwatches or sensor bands that collect physiological and movement data, including heart rate, walking patterns, and sleep activity. The collected data can be analyzed to identify abnormal patterns such as falls, prolonged inactivity, or sudden changes in physical behaviour (Patel et al., 2012).

2.2.4 Duolingo and Behavioral Gamification

Modern gamification success stories, most notably **Duolingo**, demonstrate that habit formation is driven by emotional engagement and loss aversion. According to research on Duolingo's growth strategy, the use of a persistent mascot (Duo the Owl) and a daily "streak" mechanic transforms a high-effort task into a "sticky" daily routine (Hasan, 2026). By quantifying

consistency, users feel a sense of ownership over their progress—a psychological phenomenon known as the **Endowment Effect**.

Ethernacare adapts these proven tactics by replacing standard check-in prompts with a virtual cat companion. By leveraging the emotional bond between the user and the digital pet, the system ensures high retention rates, as users are motivated to "check-in" to maintain their pet's well-being and their own reward progress.

2.3 Technology Acceptance Among the Elderly

2.3.1 Smartphone Accessibility Among Elderly Users

In recent years, smartphone ownership among older adults has increased significantly. According to the Pew Research Center (2021), smartphone adoption among individuals aged 65 and above has grown substantially compared to previous decades. The increased accessibility of smartphones provides opportunities to develop mobile-based solutions for elderly safety and monitoring. Since many elderly individuals already own smartphones, mobile applications can serve as cost-effective tools without requiring additional hardware (Pew Research Center, 2025).

2.3.2 Perceived Usefulness and Ease of Use

Recent studies show that perceived usefulness and ease of use remain strong predictors of technology adoption among older adults. Research on digital health and mobile monitoring systems indicates that elderly users are more likely to adopt applications when they believe the system improves their safety and supports independent living (Anderson et al., 2020). Confidence in smartphone usage and prior experience with digital tools also influence acceptance. When applications are perceived as complicated or unnecessary, adoption rates decrease.

2.3.3 Importance of User-Friendly Design

Usability plays a critical role in elderly technology adoption. Studies on mobile health applications emphasize that simple navigation, readable fonts, clear instructions, and minimal interaction steps significantly improve user satisfaction and long-term engagement (Dusseljee-Peute et al., 2018). Complex menus and small interface elements may discourage elderly users. Therefore, systems designed for older adults should prioritize clarity, accessibility, and reduced cognitive effort.

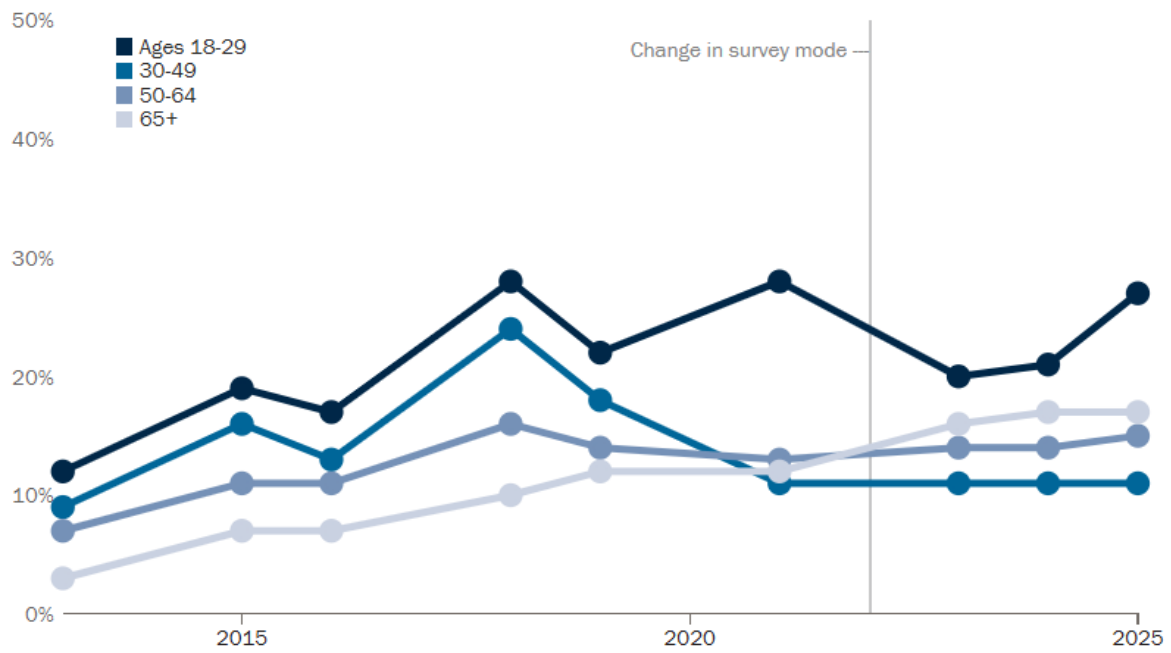


Figure 2.1: Percentages of U.S. adults who are smartphone dependent, by age

(Pew Research Center, 2025)

2.4 Similar Existing Systems in the market

2.4.1 Apple Emergency SOS

Apple Emergency SOS is a built-in smartphone feature that allows users to quickly contact emergency services and notify selected emergency contacts. The system can be activated using hardware button combinations, and it automatically shares the user's location with emergency responders and trusted contacts (Apple Inc., n.d.). The main feature is immediate emergency calling with real-time location transmission. It is integrated into iOS devices and depends on manual activation by the user.

This system is related to Ethernacare because both involve emergency alerts and contact notification. The difference is that Ethernacare uses a **daily greeting check-in mechanism** to confirm the user's status instead of requiring manual triggering. In addition, Ethernacare integrates post-death planning and digital document storage, which are not included in Apple Emergency SOS.

2.4.2 CarePredict

CarePredict is a wearable-based monitoring system designed for elderly individuals, particularly those in assisted living facilities. The system uses a wearable device equipped with sensors to monitor daily activities such as eating, sleeping, walking, and bathroom visits. It applies artificial intelligence and behavior pattern analysis to detect anomalies that may indicate health or safety risks (CarePredict, n.d.).

The main feature is continuous activity tracking through wearable sensors combined with AI-based behavioral analytics. The strength of the system lies in its detailed monitoring capabilities. However, it requires users to wear a dedicated device and is often implemented in institutional care settings.

Ethernacare shares the concept of behavior monitoring but differs by using smartphone-based passive detection rather than a wearable device. It is also designed for elderly individuals living independently at home rather than those in assisted living environments.

2.5 Limitation of the project

2.5.1 No Direct Medical Monitoring

The system does not measure heart rate, blood pressure, oxygen level, or other vital signs. It does not include fall detection using wearable sensors. It cannot diagnose medical conditions or confirm the cause of an emergency.

2.5.2 No Direct Integration with Hospitals or Emergency Services

The application does not automatically contact hospitals, ambulances, or police. Alerts are sent only to registered family members or emergency contacts. Further action depends on the response of those contacts.

2.5.3 No Smart Home or External Hardware Integration

The system does not connect to IoT devices, CCTV cameras, or home sensors. It does not require or support wearable tracking devices. Monitoring is limited to smartphone-based activity detection only.

2.6 Chapter Summary and Evaluation

This chapter analyzed the challenges of elderly isolation and the risks of delayed emergency response. It reviewed technology concepts including passive monitoring and daily check-ins, concluding that **gamification elements**—specifically interactive virtual companions—can bridge the gap between technical safety requirements and user acceptance. This review justifies the implementation of a tactile "petting" interaction as a reliable and less intrusive method of monitoring.

Chapter 3

Methodology and Requirements Analysis

3 Methodology and Requirements Analysis

This chapter describes the methodology and requirements analysis for the proposed system. It explains the development approach used in the project and identifies the stakeholders involved in the system. The chapter also presents the functional and non-functional requirements needed to support the system features. In addition, system models such as the use case diagram are introduced to illustrate how users interact with the system and how the main system functions are organized.

3.1 Methodology

3.1.1 Methodology Used: Agile

The project adopts the **Agile methodology**, which emphasizes iterative development, continuous feedback, and incremental improvements. Instead of completing the entire system at once, the system is developed in small functional modules such as daily check-in monitoring, emergency alerts, and digital planning features.

3.1.2 Methodology Phases

1. **Requirement Gathering** – Identify system requirements such as passive monitoring, alert triggering, and document storage.
2. **Design** – Create system architecture, UI wireframes, and database structure.
3. **Development** – Implement features module by module using Flutter and Supabase.
4. **Testing** – Conduct functional and usability testing for each module.
5. **Evaluation** – Review feedback and refine features before the next iteration.

3.1.3 Justification for Methodology

The Agile methodology is suitable for this project because the system consists of multiple independent modules, including activity monitoring, emergency alerts, AI chatbot guidance, and document storage. These modules can be developed and tested separately in iterative cycles. Additionally, system requirements may evolve after usability testing with potential users, especially elderly individuals, making a flexible development approach necessary. Iterative development helps reduce project risks by identifying issues early and allowing continuous improvements. It also enables critical features, such as emergency alert functions, to be tested and refined at an early stage of development.

3.2 Requirements Analysis

3.2.1 Stakeholders

The primary user is an elderly person or independent individual living alone. The user registers or signs in, completes first-login profile and consent setup, verifies phone numbers, taps Oren for daily check-ins, manages trusted contacts, configures inactivity thresholds, views rewards and weather, uses AI guidance, and records legacy-planning information.

Primary Trusted Contact

A trusted contact is an emergency recipient rather than a separate authenticated app role. The contact receives SMS follow-up when the user manually triggers an alert or reaches the configured inactivity escalation. One contact is designated as primary, and each stored contact number must be verified by SMS OTP before it can be saved.

System Administrator / Developer

The administrator or developer maintains the Flutter application, Supabase schema and policies, storage bucket, Edge Functions, OAuth providers, API secrets, logs, backups, and releases. There is currently no administrator dashboard exposed inside the end-user application.

3.2.2 Functional Requirements

Module 1: Authentication, Onboarding, and Profile

Users can register with validated email and password credentials or sign in through configured OAuth providers. New accounts complete mandatory profile details, accept the Terms and Conditions, choose Malaysian state and region values, select a blood type, configure inactivity settings, and verify the user phone number by SMS OTP. OAuth users are linked to a public profile row so that all Supabase-backed modules use the authenticated user ID consistently (Supabase, n.d.-a).

Module 2: Gamified Daily Check-In and Oren Care

Tapping Oren records at most one safety check-in per day, updates the history and streak, awards a daily token when eligible, and changes Oren mood and animation. Users can feed Oren, buy toys with Oren tokens, play with owned toys, hear interaction sounds, and return Oren to the default state after a short reaction.

Module 3: Inactivity Monitoring and Emergency Response

The configurable inactivity threshold represents one missed check-in window. Android WorkManager performs best-effort periodic checks when operating-system constraints allow (Android Developers, n.d.). The app shows reminders 1/3, 2/3, and 3/3. On the third reminder it records an alert and starts the configured escalation. Primary-contact escalation attempts SMS delivery; 999 remains a manual dialer action. A separate safe test counter sends a clearly labelled test SMS on test reminder 3.

Module 4: Trusted Contact and Location Management

Users can create, read, update, delete, and select one primary contact. Each account can store up to five contacts with validated name, relationship, international phone number, address, state, and region. Contact phone numbers require SMS OTP verification. Emergency alerts can store the device location and include a Google Maps link when location permission and positioning are available.

Module 5: Rewards and Oren Shop

The reward service calculates streaks from check-in history, synchronizes the reward catalogue, caches reward snapshots locally, displays the next reward on the dashboard, and supports virtual voucher-style rewards. Oren tokens are separate local gamification currency used for toys and interactions.

Module 6: Adaptive Malaysia Weather

The user selects a Malaysian state and region. The app requests the district or town forecast from the official data.gov.my Weather API and maps the returned Malay forecast summary to Oren backgrounds. If profile weather is unavailable, the app uses device coordinates with Open-Meteo. Successful weather responses are cached for 30 minutes (Government of Malaysia, n.d.; Open-Meteo, n.d.).

Module 7: Legacy Planning

Users can create, edit, read, and delete legacy notes; save funeral preferences; and upload or remove private PDF or image documents up to 10 MB. EternaCare stores completed documents but does not draft, validate, or notarize legal wills.

Module 8: AI Guidance

The AI Guidance screen sends the latest question and up to ten recent chat messages to a secured Supabase Edge Function. Gemini is the preferred provider, with optional OpenAI or custom-provider fallback. Conversation history is cached locally. The assistant gives general information only, does not diagnose or provide legal advice, and directs immediate Malaysian emergencies to 999 (Google AI for Developers, n.d.; Supabase, n.d.-b).

3.2.3 Functions not included

Funeral Service Booking and Payment System

While the application does not include a marketplace or transaction system. Users cannot browse specific funeral packages, service providers, or make payments for funeral services directly through the app.

Direct Communication with Service Providers

The application does not include a real-time chat or messaging system to connect users with funeral directors, lawyers, or religious organizations. The AI chatbot provides general information only and does not facilitate external business communication.

Medical and Biological Health Monitoring

The system relies solely on phone activity (screen interaction and app usage). It does not integrate with wearable devices (like smartwatches) to monitor heart rate, falls, or other biological vital signs. It is not a medical diagnostic tool.

Legal Will Drafting and Notarization

The application does not provide the legal tools to create, draft, or notarize a will. Users must obtain a legally binding will through traditional legal channels before uploading a copy to the app.

Direct Integration with Emergency Dispatch (999/911)

The system is designed to alert trusted contacts and family members first. It does not automatically dial or trigger a direct dispatch from government emergency services (Police or Ambulance) to prevent false alarms that could lead to legal liabilities or resource wastage. Except the user triggers the emergency call manually else the emergency services won't trigger.

3.2.4 Non-Functional Requirements**1. Reliability**

- On Android, inactivity checks shall use best-effort WorkManager scheduling and continue after normal app restarts when operating-system constraints permit. The system shall not claim exact-time or uninterrupted 24/7 execution.
- Emergency alerts shall include a retry mechanism if the first notification attempt fails.
- User data (contacts, preferences, documents) shall not be lost during normal app updates.
- The system shall automatically resume monitoring after device restart.

2. Usability

- The application interface shall use clear fonts and simple navigation suitable for elderly users.
- Users shall be able to configure inactivity time thresholds easily.
- The emergency status shall be clearly displayed on the home screen.

- Manual SOS and 999 actions shall use confirmation or explicit user actions. Automatic primary-contact escalation shall occur only after three missed threshold windows.

3. Performance Efficiency

- The system shall run daily reminder and check-in tracking continuously without causing noticeable device lag.
- The system shall optimize battery usage to prevent excessive power consumption during background monitoring.

3.3 Use Case Diagram

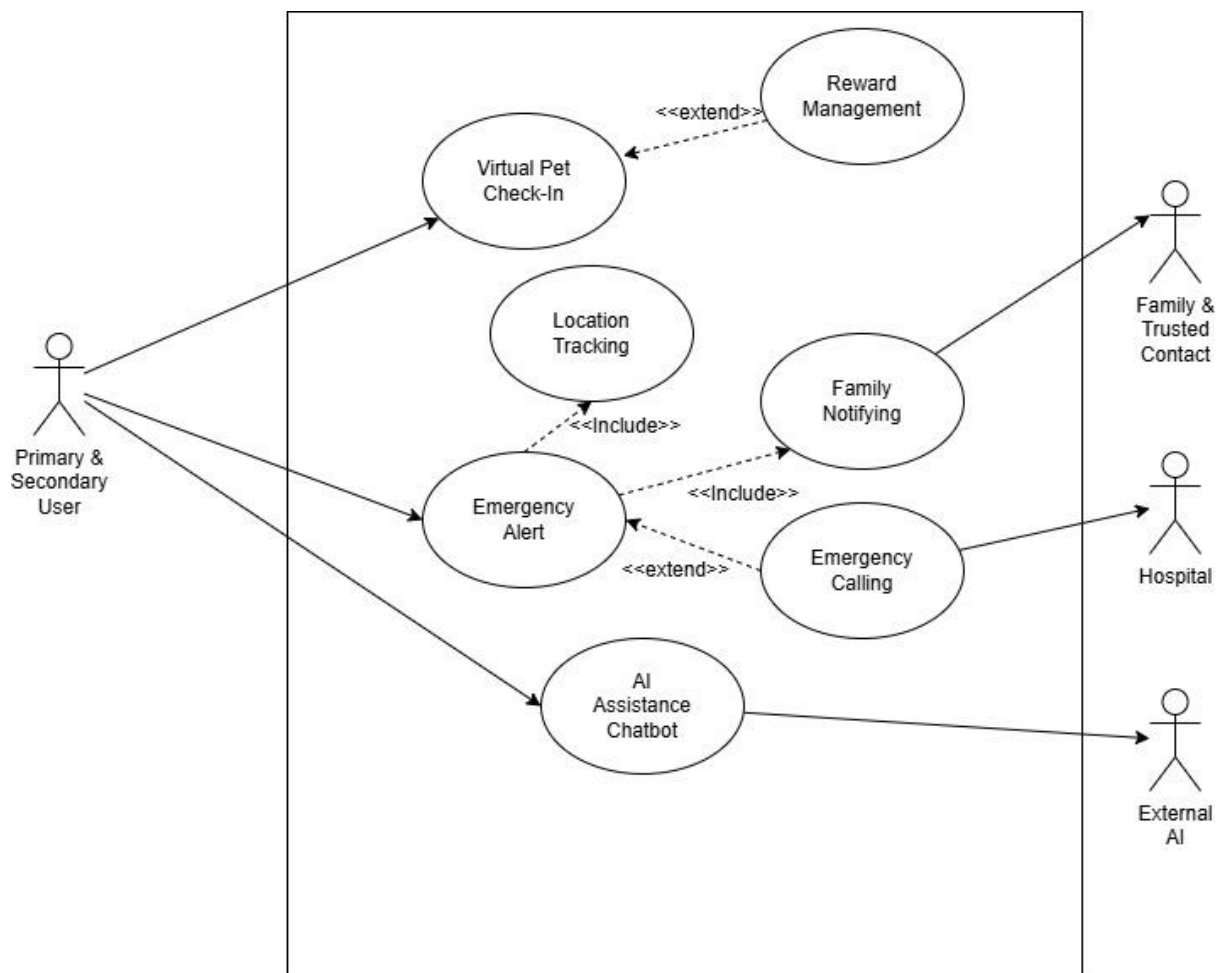


Figure 3.1: Use Case Diagram for each of stakeholders

3.3.1 Use Case Actors

Primary User - registers or signs in, completes onboarding, verifies a phone number, manages the profile and contacts, taps Oren to check in, uses Oren care and rewards, views weather, asks AI guidance, manages legacy planning, and triggers SOS.

Primary Trusted Contact - receives a verified, user-authorized emergency or test SMS. The contact does not log in to the current application.

External Services - Supabase provides authentication and protected data services; data.gov.my and Open-Meteo provide weather; Gemini provides AI content; Twilio or the Android SMS bridge provides SMS delivery; and device services provide location and notifications.

Main Use Cases

Daily Check-In - the user taps Oren; the system prevents duplicate daily records, updates history and rewards, resets the current inactivity cycle, and refreshes the dashboard.

Inactivity Escalation - the scheduler calculates elapsed threshold windows, records reminder counts, shows three notifications, and starts the configured escalation on the third reminder without automatically calling 999.

Contact Management - the user validates and verifies a contact number, stores up to five contacts, selects one primary contact atomically, and can edit or delete owned records.

Legacy Planning - the user manages funeral preferences, notes, and private documents under authenticated ownership policies.

AI Guidance - the user asks a question, receives provider or offline guidance, and retains recent local chat history.

3.4 Chapter Summary and Evaluation

This chapter detailed the Agile methodology and functional requirements for EternaCare. Key requirements were identified, focusing on the **cat-petting check-in module** as the primary safety verification mechanism. The use case diagrams illustrate how this gamified interaction integrates with the **existing rewarding system**, where consistent "pet care" (check-ins) allows users to earn physical daily necessities. These requirements form the foundation for a system that balances safety, engagement, and practical incentives.

Chapter 4

System Design

4 System Design

This chapter presents the system design of the Ethernacare application. It illustrates how the system is structured and how its components interact to fulfill the required functionalities. The chapter includes the architecture design, class diagram, entity relationship diagram (ERD), and activity diagrams to provide a clear representation of the system's structure and behavior. These design models serve as a blueprint for the system implementation and ensure that the application is well-organized, scalable, and aligned with the defined requirements.

4.1 Architecture Diagram

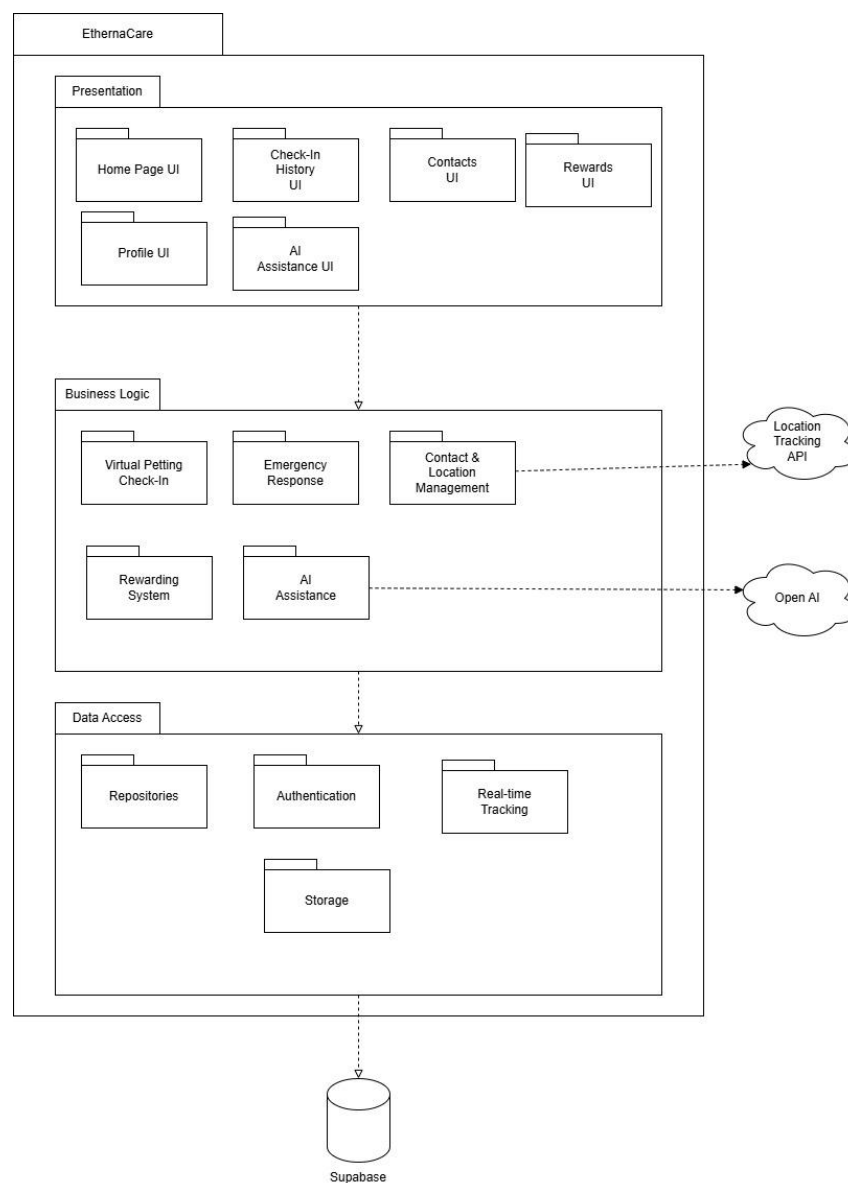


Figure 4.1: 3 Layer + 2 Tier Architecture diagram

4.1.1 Architecture Justification

EthernaCare follows a layered Flutter client and cloud-service architecture. The Presentation layer contains responsive screens and reusable widgets. Controllers and services coordinate validation, check-ins, inactivity policy, rewards, weather, AI, notifications, SMS, Oren care, and local caching. Repository classes isolate Supabase Auth, PostgreSQL, Storage, and Edge Function access. Models provide typed serialization, while Android background components invoke the inactivity service outside the foreground UI. This separation follows Flutter guidance that distinguishes UI responsibilities from repositories and services in the data layer (Flutter, n.d.).

Presentation Layer

The Presentation layer is responsible for handling all user interactions through interfaces such as the Home Page UI, Check-In History UI, Contacts UI, Rewards UI, Profile UI, and AI Assistance UI. This layer is designed to provide a simple, clear, and user-friendly experience, which is especially important for elderly users who may not be familiar with complex applications. By separating the user interface from the system logic, changes to the UI can be made without affecting the core functionality of the system. This improves usability, maintainability, and allows for easier future enhancements such as redesigning the interface or adding new features.

Business Logic Layer

The business and service layer applies application rules, including one check-in per day, reward synchronization, contact validation, primary-contact selection, three-stage inactivity escalation, safe test flows, weather caching, AI fallback behavior, and Oren state transitions. Controllers remain thin and delegate reusable work to services.

Data Access Layer

The Data Access layer contains repositories for authentication, users, check-ins, contacts, rewards, emergencies, legacy planning, and documents. Repositories use the authenticated Supabase session and RLS-protected tables. Edge Functions hold third-party secrets and integrate Gemini, Twilio, and phone OTP delivery without exposing provider credentials in the Flutter client (Supabase, n.d.-b; Supabase, n.d.-c).

4.1.2 Architecture Advantages

- **Scalability:** Individual layers in the architecture can be scaled independently to meet performance needs.
- **Flexibility:** Different technologies can be used within each layer without affecting others.
- **Maintainability:** Changes in one layer do not necessarily impact other layers, thus simplifying the maintenance. (Geeksforgeeks, 2025)

4.2 Activity Diagram

4.2.1 Login Process

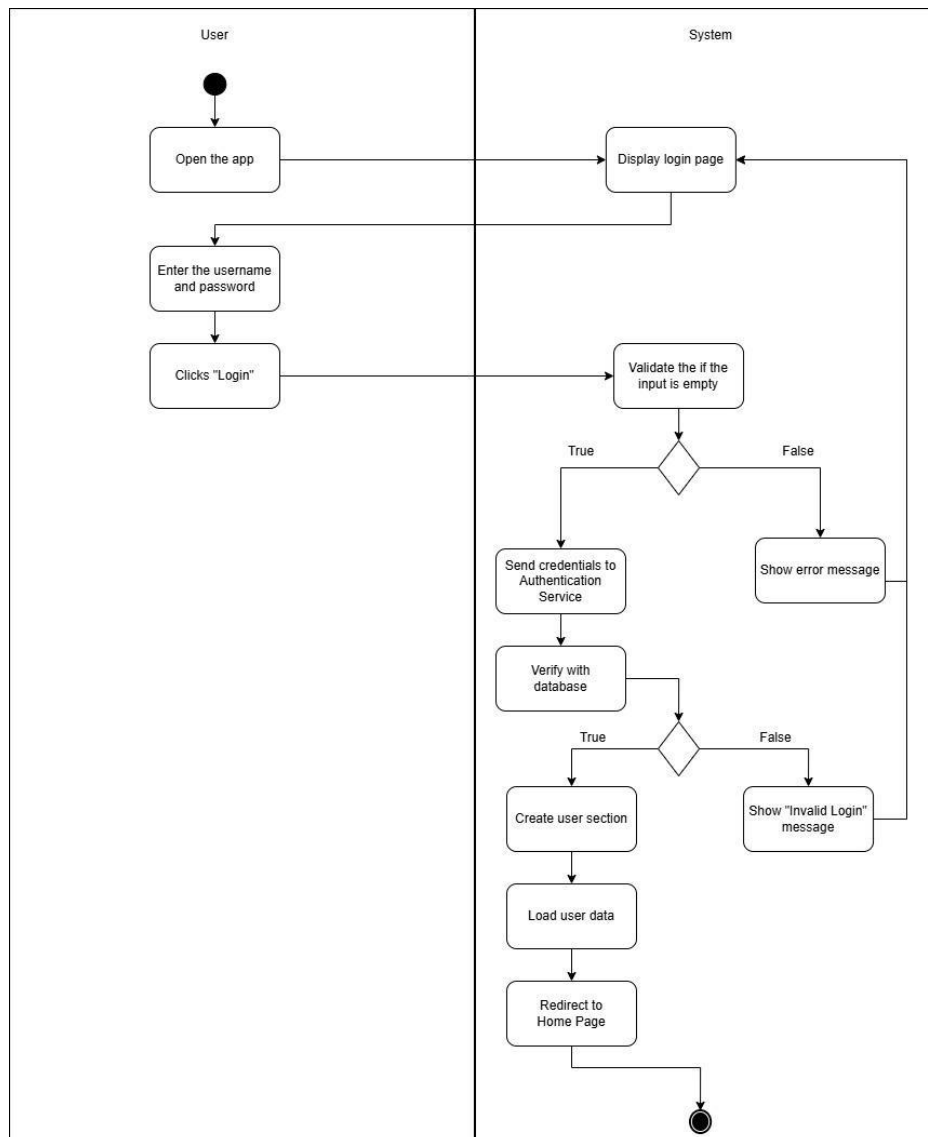


Figure 4.2.1: Activity Diagram show the login process of the Ethernacare

The login flow validates email and password inputs or starts a configured OAuth provider. Supabase creates a JWT-backed session and the app ensures that an associated public user profile exists. A new account is routed through mandatory profile, Terms and Conditions, user-phone OTP, and primary-contact OTP setup before reaching the dashboard. Returning users with complete setup proceed directly to the home screen. Friendly error dialogs replace raw provider exceptions.

4.2.2 Virtual Petting Check-In

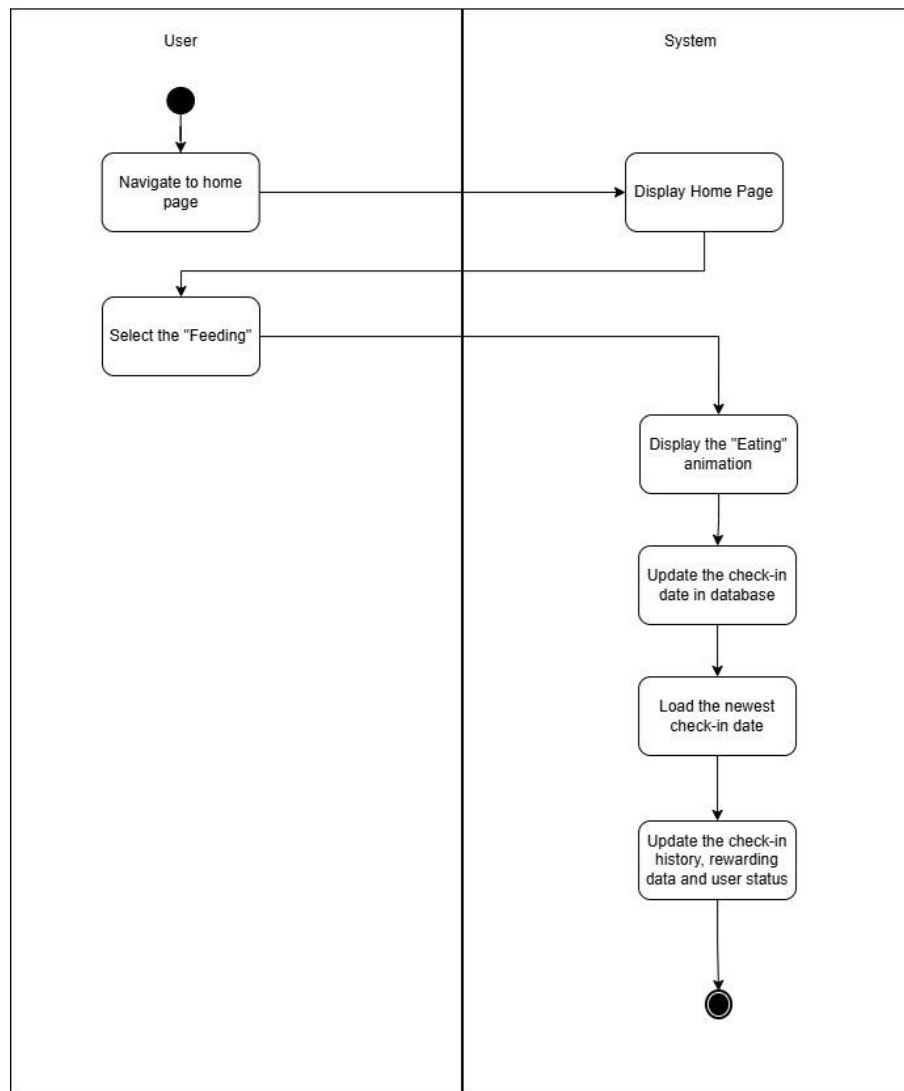


Figure 4.2.2: Activity Diagram show the process of Virtual Petting

The user taps Oren directly rather than pressing a separate Pet button. The repository checks for an existing record in the current local day and inserts a new Supabase check-in only when one does not exist. The service then refreshes the dashboard, streak, rewards, Oren tokens, mood, sound, and interaction animation. A completed check-in also makes any earlier local inactivity reminder state obsolete.

4.2.3 Emergency Alert

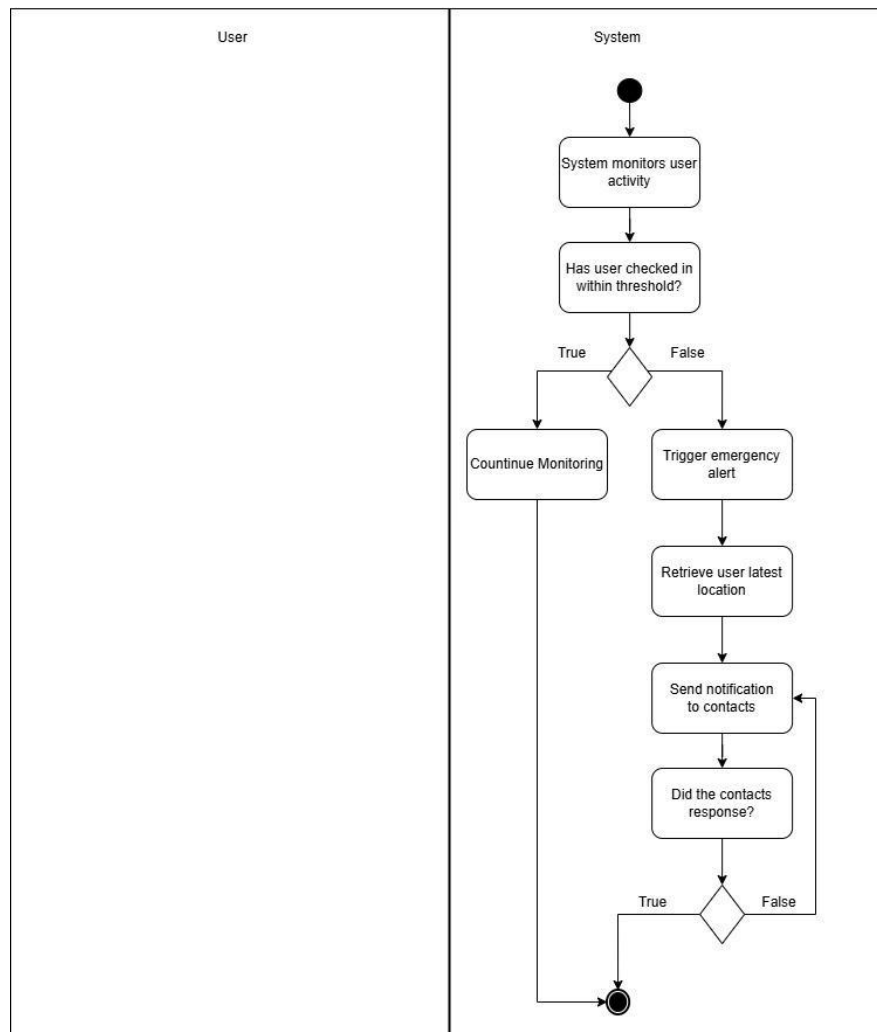


Figure 4.2.3: Activity Diagram show the process of Emergency Alert

The emergency process supports manual SOS and automatic inactivity escalation. For inactivity, the service divides elapsed time since the latest check-in by the user's configured threshold. It shows one local notification per missed window and records a visible 0/3 to 3/3 dashboard counter. Reminder 3 starts escalation. If the profile target is the primary trusted contact, the app retrieves that contact, records an emergency alert, attempts Android SMS, and otherwise queues a Twilio Edge Function delivery with retry information. Location is attached when available. If the user selects 999, the inactivity event records a critical notice but does not automatically dial emergency services. Test alerts are stored separately and cannot suppress or duplicate real alerts.

4.3 ERD Relationship Diagram

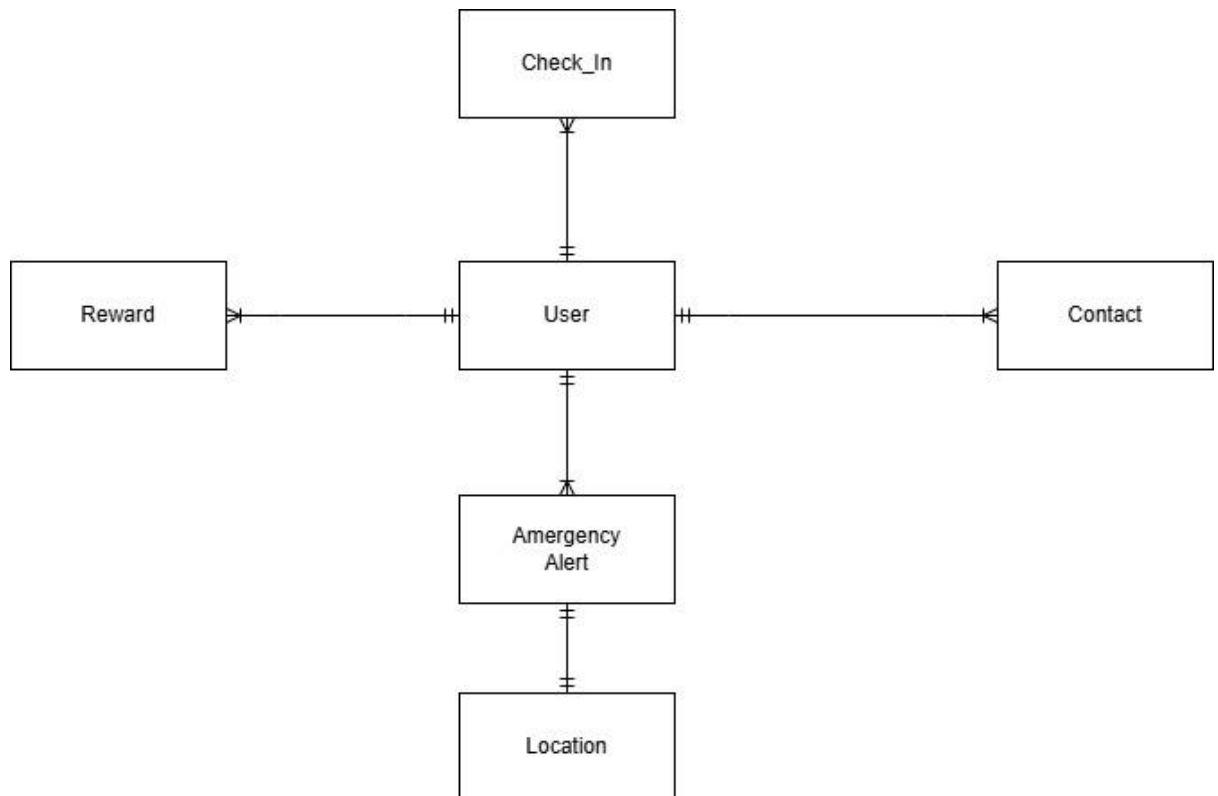


Figure 4.3: ERD diagram that describes the relationship between the entities

4.3.1 Entities attributes

User

Attributes: id, name, email identity, phone, phone_verified_at, address, address_state, address_region, blood_type, inactivity_threshold, emergency_escalation_target, terms_version, terms_accepted_at, and profile_completed_at. Age is not collected.

Contact

Attributes: id, user_id, name, relationship, phone, phone_verified_at, address, address_state, address_region, and is_primary. One user owns up to five contacts and one primary contact.

Check-In

Attributes: id, user_id, checkin_time, and status. A user owns many daily activity records.

Emergency Alert and Location

Emergency alerts contain id, user_id, triggered_time, and status. Optional location records contain alert_id, latitude, longitude, and timestamp.

Reward and Reward Catalogue

User rewards contain streak, reward type/code, status, earned and redeemed timestamps. The catalogue stores title, sponsor, milestone, kind, voucher value, version, and active state.

Legacy Planning

Funeral preferences are one-to-one with the user. Legacy notes and secure documents are one-to-many. Documents store private storage paths and upload timestamps.

SMS Delivery and Phone Verification

The emergency delivery outbox stores contact, message, provider, attempts, status, errors, and processed time. OTP records store only a code hash, purpose, expiry, consumption state, and attempt count; successful verification records are retained separately.

4.3.2 ERD Relationships

1. User → Contact (1:M)

One user can have multiple contacts, but each contact belongs to only one user. This allows multiple family members or trusted contacts to be notified during emergencies.

2. User → Check-In (1:M)

One user can perform many check-ins over time. Each check-in records user activity to detect inactivity and ensure safety monitoring.

3. User → Emergency Alert (1:M)

One user can trigger multiple emergency alerts. Each alert represents a separate incident and is stored for tracking purposes.

4. Emergency Alert → Location (1:1)

Each emergency alert is linked to one location record. This ensures accurate location tracking during emergencies.

5. User → Reward (1:M)

One user can earn multiple rewards. This supports the gamification system based on consistent check-ins.

6. User → Document (1:M)

One user can store multiple documents (e.g., will, funeral preferences). Each document belongs to a single user to maintain privacy and ownership.

4.4 Security Handling

Authentication and Access Control

Supabase Auth supports email/password and configured OAuth providers. Authenticated sessions use JWTs. A database trigger creates the matching public profile for email and OAuth accounts so every repository uses the same authenticated user ID (Supabase, n.d.-a).

Row Level Security and Private Storage

RLS is enabled on exposed user-owned tables. Policies compare `auth.uid()` with `id` or `user_id` for select, insert, update, and delete operations. The legacy-documents bucket is private and storage policies restrict paths to the authenticated user folder (Supabase, n.d.-c).

Phone Verification and Input Validation

Six-digit OTP codes are generated server-side, stored as hashes with expiry and attempt limits, and delivered through an Edge Function. Database triggers reject changed profile or contact phone numbers that do not have a recent successful verification. Client and database validation also enforce name, password, phone, address, contact-count, and primary-contact rules.

Secrets and Secure Communication

The Flutter client communicates through HTTPS. Gemini, Twilio, SMS worker, and OTP secrets are stored as Supabase Edge Function environment secrets rather than in source code. Edge Functions validate authenticated users or a worker secret before privileged operations (Supabase, n.d.-b).

4.5 UI Prototype

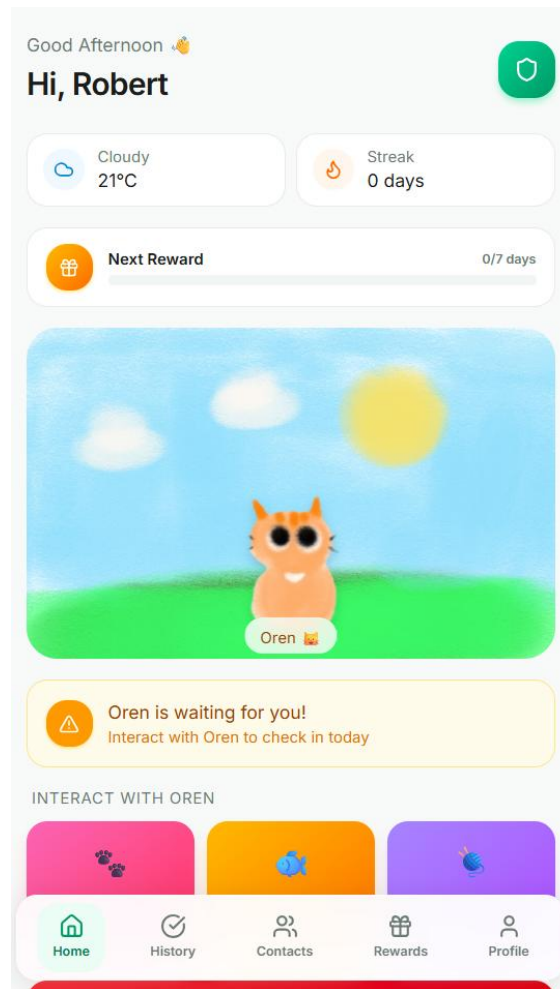


Figure 4.4.1: Home Page

The current home page uses a responsive premium dashboard. Oren is the direct check-in target and changes pixel-art mood, energy, interaction, and weather background. The page also shows Malaysian region weather, streak and check-in totals, the next reward, real inactivity reminder progress, a separate three-step test button, emergency state, Oren status, shop controls, SOS, SMS testing, and sign-out. Scroll cues and safe-area spacing support small Android screens.

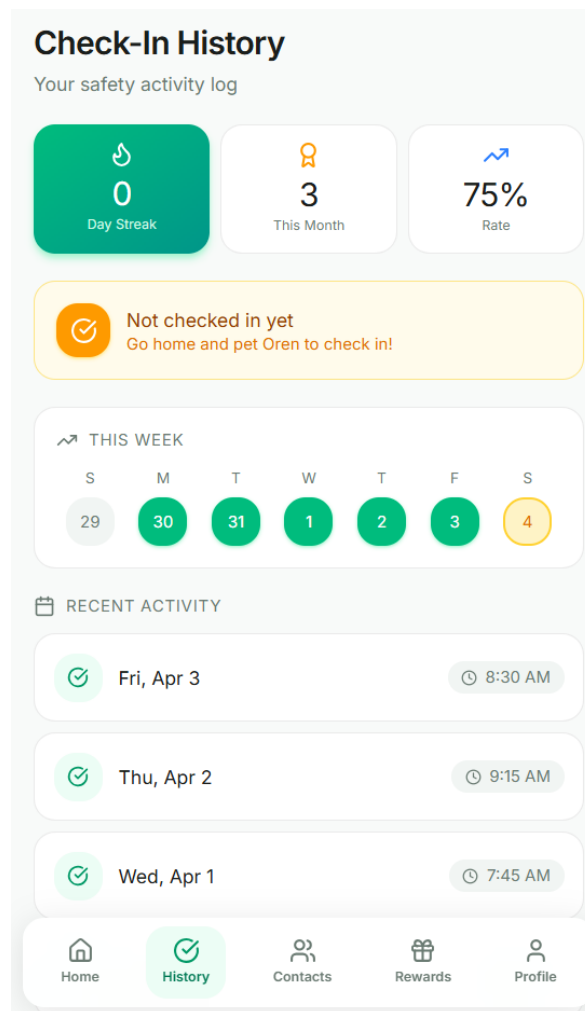


Figure 4.4.2: Check-In History Page

The Check-in History page provides a transparent and structured record of the user's daily interactions, serving as a visual "health diary" of their activity. It presents a chronological list of every successful "petting" event, displaying specific dates and timestamps to confirm when the "heartbeat signal" was sent to the system. This interface allows both the user and their caregivers to monitor consistency, highlighting daily streaks and any missed days that might indicate a change in routine. By centralizing this data, the page reinforces the habit-forming nature of the gamified system while offering peace of mind through a clear, verifiable history of the user's well-being and safety status.

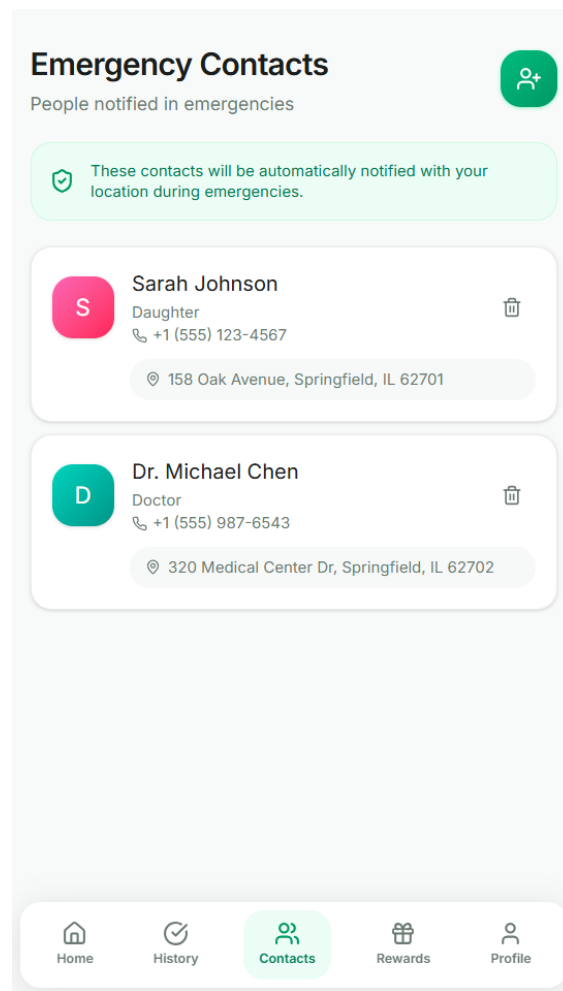


Figure 4.4.3: Emergency Contacts Page

The Emergency Contacts page lists up to five owned contacts with name, relationship, verified international phone number, address, state, and region. A star action changes the primary contact through an atomic database function, while edit and delete operations refresh the interface immediately. Adding or changing a phone number requires SMS OTP verification.

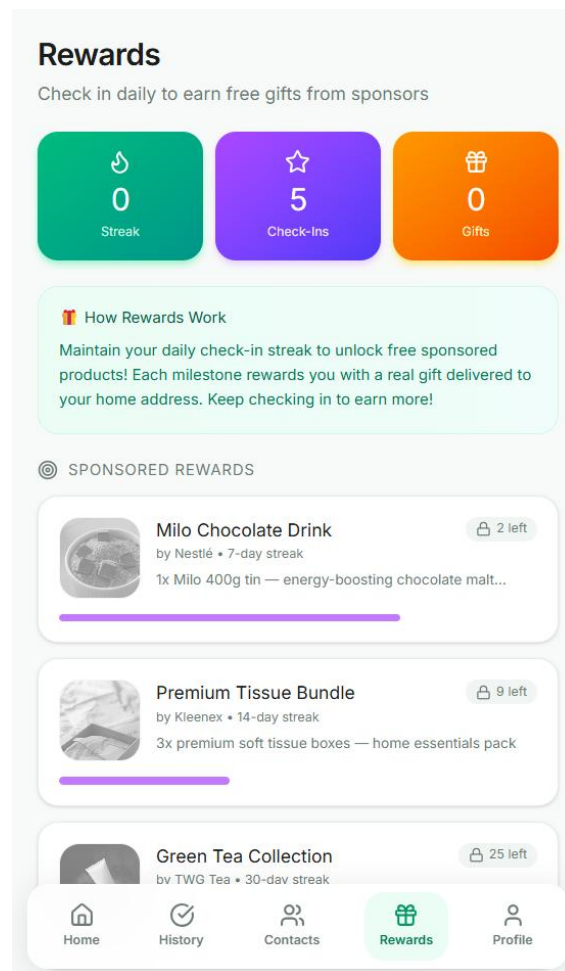
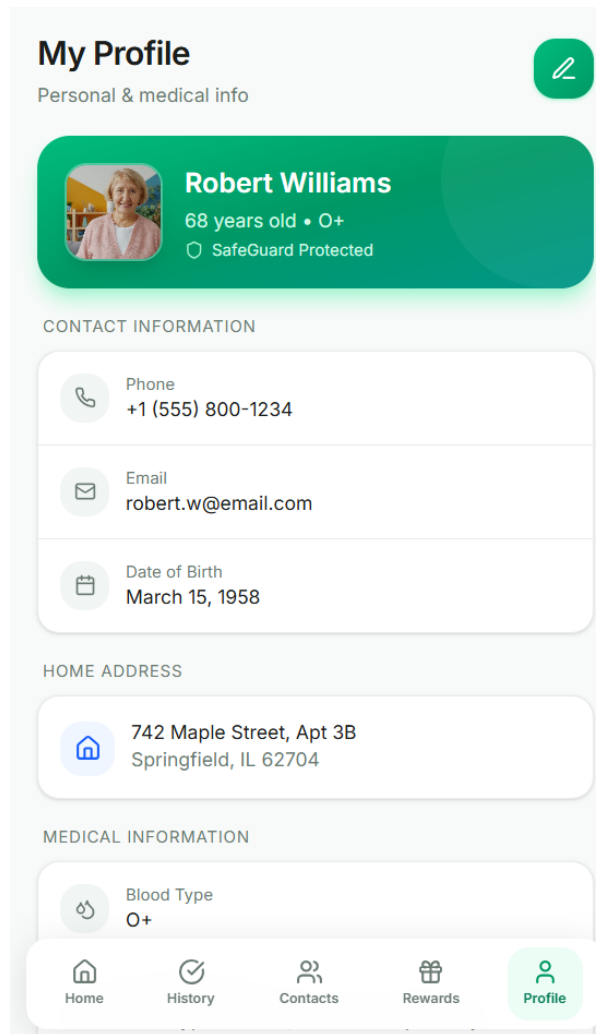


Figure 4.4.4: Rewards Page

The Rewards page combines server-backed streak rewards and locally cached reward catalogue data with Oren tokens and toys. The dashboard shows the next milestone reward, while the Oren shop allows owned toys to be purchased and used. Reward examples include virtual sponsor vouchers rather than guaranteed physical inventory.



My Profile
Personal & medical info

Robert Williams
68 years old • O+
SafeGuard Protected

CONTACT INFORMATION

Phone
+1 (555) 800-1234

Email
robert.w@email.com

Date of Birth
March 15, 1958

HOME ADDRESS

742 Maple Street, Apt 3B
Springfield, IL 62704

MEDICAL INFORMATION

Blood Type
O+

Home History Contacts Rewards Profile

Figure 4.4.5: Profile Page

The Profile page displays name, verified phone, home address, Malaysian state and region, blood type, inactivity threshold, and escalation target. Age has been removed. Editing uses a full-page responsive form, country calling-code selection, validation, blood-type selection, and OTP verification when the phone number changes. The page also links to AI Guidance and Legacy Planning.

4.6 Chapter Summary and Evaluation

This chapter presented the overall system design of the Ethernacare application, including the architecture, diagrams, and security considerations. The system adopts a three-tier architecture consisting of the Presentation, Business Logic, and Data Access layers to ensure a clear separation of concerns and improve maintainability. Various design models such as the class diagram, entity relationship diagram (ERD), and activity diagrams were developed to illustrate the system structure, data relationships, and process flows, including login and emergency alert functionalities. In addition, appropriate design patterns and security handling techniques were incorporated to enhance system reliability, scalability, and data protection. Overall, this chapter provides a comprehensive blueprint that guides the implementation of the system in the next phase.

Chapter 5

Implementation and Testing

5 Implementation and Testing

Implementation Overview

The application was implemented as a Flutter project using a layered structure: presentation screens and widgets, controllers, domain services, repositories, typed models, background components, and utility validation. Supabase provides the remote backend, while SharedPreferences provides lightweight device-local JSON caching. The repository contains SQL migrations and audits so the deployed schema can be checked against the application model.

Authentication and Data Protection

Email registration, sign-in, resend verification, manual signup-code verification, and OAuth are encapsulated in AuthRepository. AuthGate checks session, profile completion, accepted terms, primary contact, and tutorial state. Supabase RLS protects each user-owned table, while private storage policies protect legal documents. Phone OTP request and verification logic is isolated in two Edge Functions.

```
create policy "contacts_select_own" on public.contacts for select
to authenticated using ((select auth.uid()) = user_id);
```

Check-In, Rewards, and Local Caching

CheckinRepository prevents more than one record per local calendar day. CheckinService refreshes cached history and RewardService calculates unique consecutive dates. OrenCareService persists tokens, owned toys, energy, mood, and daily bonus dates locally. Dashboard, reward, weather, chat, and inactivity snapshots are read from cache first and refreshed from the server when needed.

Inactivity Monitoring and Emergency SMS

A 15-minute Android WorkManager task initializes Supabase in a background isolate and calls InactivityService. The service derives the number of missed windows from elapsed time, stores reminder state locally, suppresses duplicate alerts for the same check-in, and escalates at three reminders. EmergencyService records the alert, obtains location when permitted, attempts direct Android SMS, creates an outbox row when needed, and invokes the Twilio worker. The outbox records attempts, provider IDs, errors, and completion state (Android Developers, n.d.; Twilio, n.d.).

```
missedCheckIns = elapsedSeconds ~/ thresholdSeconds;
if (missedCheckIns >= 3) triggerEmergencyDetailed(sendAutomatedSms: true);
```

Weather, AI, and Legacy Planning

WeatherService queries the selected Malaysian region through data.gov.my and falls back to Open-Meteo coordinates. Forecast text maps to clear, cloudy, rain, or thunderstorm Oren scenes. AiService invokes a server-side Gemini Edge Function with limited conversation history and a safety-constrained prompt, then falls back to built-in guidance if unavailable. LegacyPlanningRepository provides CRUD for preferences, notes, document metadata, and private storage.

Testing Approach

Automated Flutter logic and widget tests cover models, validation, responsive forms, Oren interactions, weather mapping, rewards, AI history, and inactivity boundaries. The current suite contains 26 passing tests. Provider-dependent OAuth, SMS, OTP, AI, weather, location, and background execution additionally require device and deployed-environment integration tests.

ID	Test case	Expected result	Status
TC01	Email/OAuth authentication UI and validation	Valid input creates or restores a session and profile route	Pass
TC02	First-login profile, terms, and phone verification	Incomplete setup is blocked; verified setup continues	Pass
TC03	Tap Oren daily check-in	One daily record, refreshed history, token and streak update	Pass
TC04	Contact CRUD and primary selection	Validated contact updates immediately; one primary contact	Pass
TC05	Reward and local cache serialization	Streak, catalogue, tokens, toys, and snapshots restore	Pass
TC06	Malaysia weather mapping	Clear, rain, thunderstorm, and night scenes map correctly	Pass
TC07	Legacy planning models and CRUD UI	Preferences, notes, and documents serialize correctly	Pass
TC08	AI history and offline guidance	Recent history restores and safe fallback answers are available	Pass
TC09	Three-stage inactivity policy	24/48/72-hour boundaries calculate 1/2/3 misses	Pass
TC10	Reminder test cycle	Third safe test trigger starts labelled test SMS flow	Pass
TC11	External provider delivery	OAuth, OTP, Twilio, Gemini, weather, and location	Integration setup required

Implementation Evaluation

The implementation satisfies the principal requirements while keeping high-risk actions explicit. Raw provider errors are converted into user-facing dialogs, local caching reduces repeated requests, test alerts are separated from real alerts, and 999 cannot be contacted automatically. Remaining operational reliability depends on correct Supabase migrations, Edge Function deployment, provider secrets, Android permissions, network availability, and operating-system scheduling.

5.1 Chapter Summary and Evaluation

This chapter described the implemented Flutter layers, Supabase security, cached data strategy, background inactivity monitoring, emergency delivery, weather, AI, legacy planning, and automated tests. The completed modules demonstrate that the design has been translated into working, testable components while external-provider dependencies remain clearly identified.

Chapter 6

System Deployment

6 System Deployment

Deployment Scope and Requirements

EternaCare is deployed as a Flutter Android application with optional web testing. Development requires Flutter/Dart, Android Studio or the Android SDK, and a Supabase project. Android devices require internet, notification and location permissions; direct-device SMS additionally requires SEND_SMS permission. Production SMS should use the server-side Twilio worker.

Backend and API Setup

The deployment process runs the schema repair or migration scripts, validation constraints, primary-contact RPC, RLS policies, legacy-planning/storage setup, SMS outbox setup, OAuth profile bootstrap, and phone OTP verification SQL. Edge Functions for AI guidance, emergency SMS, OTP request, and OTP verification are deployed with the Supabase CLI. Gemini, Twilio, OTP, and worker credentials are stored as project secrets rather than committed to source control.

Android Build and Installation

Dependencies are restored with flutter pub get. Automated tests are run before flutter build apk or an Android Studio release build. OAuth redirect URIs and Android signing fingerprints must match provider configuration. The APK or Play release is installed, permissions are granted, and email/OAuth, check-in, contact, weather, AI, location, OTP, and SMS flows are tested on a physical device.

Backup and Risk Management

Source code and SQL migrations are version-controlled. Supabase database and Storage backups should follow the selected project plan. Secrets must be rotated if exposed. External-provider quotas and failures are mitigated with offline AI guidance, cached weather and dashboard data, SMS outbox retries, clear error dialogs, and safe manual 999 guidance. WorkManager timing is best-effort and should not be described as exact-time emergency dispatch.

Training and Follow-Up

First-login onboarding and the in-app tutorial explain Oren check-in, contacts, rewards, SOS, and profile settings. Deployment verification should include a supervised three-reminder test using a consented primary contact. Logs, failed outbox rows, user feedback, crash reports, API quotas, and schema audits should be reviewed after release.

Chapter Summary and Evaluation

This chapter defined the practical deployment sequence, permissions, backend migrations, Edge Functions, secrets, backup controls, training, and follow-up checks. It also distinguishes tested application behavior from external-provider and operating-system guarantees.

Discussions and Conclusion

7 Discussions and Conclusion

Summary

EternaCare addresses delayed awareness of prolonged inactivity through a low-effort daily Oren interaction. Flutter provides a responsive interface, Supabase provides authenticated and RLS-protected cloud data, SharedPreferences improves perceived performance, and external APIs add weather, AI, location, and SMS capabilities. Agile iteration was appropriate because the project evolved through authentication, schema, contact, onboarding, responsiveness, and emergency-flow refinements.

Achievements

The project implements all principal modules: email and OAuth authentication, first-login consent and verified profile setup, Oren check-ins and interactions, check-in history, trusted-contact CRUD and primary selection, rewards and Oren shop, Malaysian weather backgrounds, emergency location records, three-stage inactivity escalation, safe SMS testing, AI guidance with history and offline fallback, and legacy-planning CRUD. The automated Flutter suite reports 26 passing tests.

Contributions

The main contribution is the combination of a friendly virtual-pet habit with an explicit safety signal and measured escalation. Oren makes a repetitive check-in less clinical, while the dashboard keeps safety status visible. Local-first reads, test-only emergency paths, verified contact numbers, and separation between trusted-contact SMS and manual 999 calling improve usability and reduce avoidable risk.

Limitations and Future Improvements

Android background execution is best-effort and can be delayed by battery policy or force-stop behavior; web has no periodic background scheduler. SMS, OTP, OAuth, Gemini, and weather depend on external configuration, availability, quotas, and billing. Local cache is device-specific. EternaCare is not a medical monitor and does not detect falls or vital signs. Future work should add server-scheduled monitoring, push notifications, delivery receipts, stronger integration tests, accessibility and elderly-user studies, multilingual content, and an authorized caregiver portal.

Issues and Solutions

OAuth accounts initially lacked matching public profiles, so an authentication trigger and backfill were added. Database schema-cache errors were addressed with auditable idempotent SQL migrations. Primary-contact conflicts were resolved through a unique partial index and atomic RPC. Contact forms were moved from small dialogs to responsive pages. Blank post-login routing was isolated from the tutorial and guarded by explicit setup states. Weather false-rain behavior was corrected by prioritizing the selected Malaysia region and mapping official forecast summaries. Raw asynchronous and provider exceptions were replaced with friendly error dialogs, while automated tests protect the repaired flows.

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Appendices

APPENDIX A User Guide

Account and First Login

Register with a valid email and strong password or use a configured OAuth provider. Complete required profile details, accept the Terms and Conditions, verify the user phone, and add and verify a primary trusted contact.

Daily Use

Open Home and tap Oren once each day to check in. Review History for timestamps, Rewards for streak milestones, and the dashboard for weather, next reward, and inactivity state. Use Feed and owned toys for Oren interactions.

Safety Functions

Maintain an accurate primary contact and inactivity threshold. The real counter reaches three reminders before configured escalation. Use the Safe reminder test only with the contact's consent. For immediate danger, explicitly open the 999 dialer and place the call.

Profile, AI, and Legacy Planning

Edit state, region, blood type, threshold, and escalation target from Profile. AI Guidance provides general information only. Legacy Planning stores preferences, notes, and completed private documents; it does not create legal documents.

APPENDIX B Developer Guide

Development Requirements

Install Flutter/Dart, Android Studio or Android SDK tools, Git, and Node.js for npx-based Supabase CLI commands. Restore packages with flutter pub get and verify with flutter test.

Supabase Setup

Create the Supabase project, configure the app URL and anon key, run the SQL migration and RLS files, create the private legacy-documents bucket, configure OAuth redirect URLs, deploy Edge Functions, and set Gemini, Twilio, OTP, and worker secrets. Never commit service-role or provider secrets.

Build and Verification

Run `schema_audit.sql`, execute the Flutter tests, build an Android APK or release bundle, install on a physical device, grant permissions, and test email/OAuth, OTP, contacts, check-in, rewards, weather, AI, legacy documents, location, notifications, and the safe three-reminder SMS cycle.

Maintenance

Monitor Supabase Auth, database, Storage, Edge Function logs, SMS outbox failures, API quotas, and crash reports. Apply idempotent migrations before releasing code that expects new columns. Rotate leaked credentials immediately and keep provider billing and sender verification current.